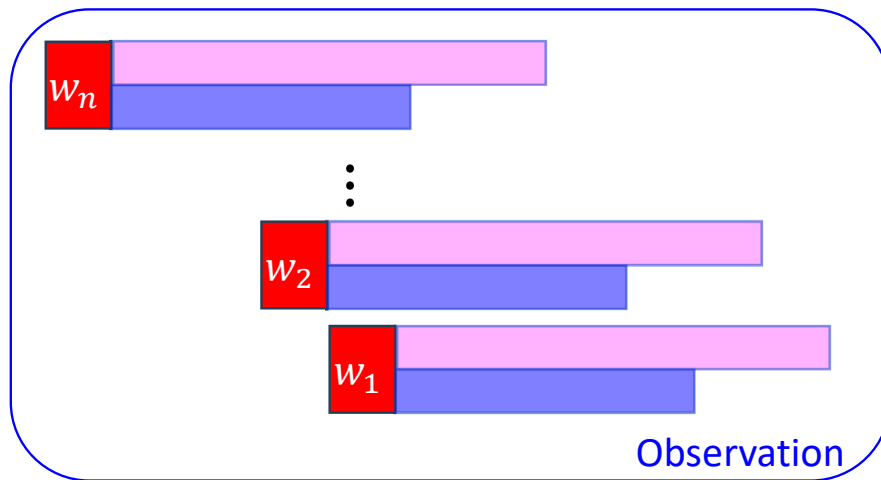
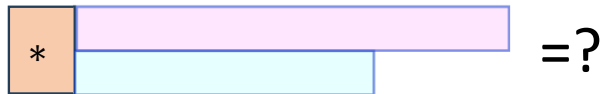


ARMA: Forecast

- MMSE linear predictor
- Durbin-Levinson algorithm
- Iterative algorithm

Forecasting model



$$\hat{x}_{n+m} = f(x_1, x_2, \dots, x_n)$$

Objective: Find $\hat{x}$ that produces MSE.

$$x_{n+m}^n = \hat{x}_{n+m} = E(x_{n+m} | x_1, x_2, \dots, x_n)$$

For simplicity, consider linear estimator.

$$x_{n+m}^n = \overset{\text{Mean} = 0}{\cancel{\alpha_0}} + \sum_{k=1}^n \alpha_k x_k$$

MMSE linear estimator

$$\text{MSE} = E[x_{n+m} - \boxed{x_{n+m}^n}]^2 = E \left[x_{n+m} - \sum_{k=1}^n \alpha_k x_k \right]^2$$

$x_{n+m}^n = \sum_{k=1}^n \alpha_k x_k$

Take the partial derivative and set the derivative to 0

$$\frac{\partial \text{MSE}}{\partial \alpha_l} = E \left[2 \left[x_{n+m} - \sum_{k=1}^n \alpha_k x_k \right] (-x_l) \right]$$

$$0 = E[x_{n+m} x_l] - \sum_{k=1}^n \alpha_k E[x_k x_l] = \gamma(n+m-l) - \sum_{k=1}^n \alpha_k \gamma(k-l)$$

$$\gamma(n+m-l) = \sum_{k=1}^n \gamma(k-l) \alpha_k, \quad l = 1, 2, \dots, n$$

α_k

l

Unknown

MMSE linear estimator (2)

$$\sum_{k=1}^n \gamma(k-l) \alpha_k = \gamma(n+m-l), \quad l = 1, 2, \dots, n$$

Convert this linear system to matrix form

$$\gamma(0)\alpha_1 + \gamma(1)\alpha_2 + \dots + \gamma(n-1)\alpha_n = \gamma(n+m-1)$$

$$\gamma(1)\alpha_1 + \gamma(0)\alpha_2 + \dots + \gamma(n-2)\alpha_n = \gamma(n+m-2)$$

$$\vdots$$

$$\gamma(n-1)\alpha_1 + \gamma(n-2)\alpha_2 + \dots + \gamma(0)\alpha_n = \gamma(m)$$

MMSE linear estimator (3)

Covariance matrix: $E(\mathbf{x}\mathbf{x}')$

$$\begin{bmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(n-2) \\ \gamma(2) & \gamma(1) & \cdots & \gamma(n-3) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(n-1) & \gamma(n-2) & \cdots & \gamma(0) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \vdots \\ \alpha_n \end{bmatrix} = \begin{bmatrix} \gamma(n+m-1) \\ \gamma(n+m-2) \\ \gamma(n+m-3) \\ \vdots \\ \gamma(m) \end{bmatrix}$$

$$x_{n+m}^n = \sum_{k=1}^n \alpha_k x_k = \mathbf{A}'_n \mathbf{x}$$

$$\Gamma_n \mathbf{A}_n = \boldsymbol{\gamma}_n^{(m)}$$

$$\mathbf{A}_n = \Gamma_n^{-1} \boldsymbol{\gamma}_n^{(m)}$$

$$\text{MSE} = E[x_{n+m} - \mathbf{A}'_n \mathbf{x}]^2$$

Mean square prediction error

Prediction

Mean square prediction error

$$P_{n+m}^n = MSE = E[x_{n+m} - \mathbf{A}'_n \mathbf{x}]^2$$

$$= E[x_{n+m}^2 - 2x_{n+m} \mathbf{A}'_n \mathbf{x} + (\mathbf{A}'_n \mathbf{x})^2]$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Scalar value: $(\mathbf{A}'_n \mathbf{x}) = (\mathbf{A}'_n \mathbf{x})'$

$$= E[x_{n+m}^2] - 2\mathbf{A}'_n E \begin{bmatrix} x_1 x_{n+m} \\ x_2 x_{n+m} \\ \vdots \\ x_n x_{n+m} \end{bmatrix} + E[(\mathbf{A}'_n \mathbf{x})(\mathbf{A}'_n \mathbf{x})]$$

Covariance matrix Γ_n

$$= E[x_n^2] - 2\mathbf{A}'_n \begin{bmatrix} \gamma(n+m-1) \\ \gamma(n+m-2) \\ \vdots \\ \gamma(m) \end{bmatrix} + E[(\mathbf{A}'_n \mathbf{x})(\mathbf{x}' \mathbf{A}_n)]$$

$$= \gamma(0) - 2\mathbf{A}'_n \boldsymbol{\gamma}_n^{(m)} + \mathbf{A}'_n \boldsymbol{\Gamma}_n \mathbf{A}_n = \gamma(0) - 2\mathbf{A}'_n \boldsymbol{\gamma}_n^{(m)} + \mathbf{A}'_n \boldsymbol{\Gamma}_n \boldsymbol{\Gamma}_n^{-1} \boldsymbol{\gamma}_n^{(m)}$$

Mean square prediction error (2)

$$\begin{aligned}P_{n+m}^n &= \gamma(0) - 2\boldsymbol{\gamma}_n^{(m)'} \Gamma_n^{-1} \boldsymbol{\gamma}_n^{(m)} + \boldsymbol{\gamma}_n^{(m)'} \Gamma_n^{-1} \Gamma_n \Gamma_n^{-1} \boldsymbol{\gamma}_n^{(m)} \\&= \gamma(0) - \boldsymbol{\gamma}_n^{(m)'} \Gamma_n^{-1} \boldsymbol{\gamma}_n^{(m)}\end{aligned}$$

Consider one-step-ahead predictor ($m = 1$) and let $\boldsymbol{\gamma}_n^{(m)} = \boldsymbol{\gamma}_n$

$$\begin{aligned}P_{n+1}^n &= \gamma(0) - \boldsymbol{\gamma}_n^{(m)'} \Gamma_n^{-1} \boldsymbol{\gamma}_n^{(m)} \\&= \gamma(0) - \boldsymbol{\gamma}_n' \Gamma_n^{-1} \boldsymbol{\gamma}_n\end{aligned}$$

Reduction of MSE from the known information.

$\gamma(0) = E[x - E(x)]^2 = \text{MSE when the prediction is the mean of } x.$

One-step-ahead predictor

$$x_{n+1}^n = \sum_{k=1}^n \alpha_k x_k$$

Mapping information from x_k to x_{n+1} → $\phi_{n,(n+1-k)}$

Covariance matrix: Γ_n

$$\begin{bmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(n-2) \\ \gamma(2) & \gamma(1) & \cdots & \gamma(n-3) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(n-1) & \gamma(n-2) & \cdots & \gamma(0) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \vdots \\ \alpha_n \end{bmatrix} = \begin{bmatrix} \gamma(n) \\ \gamma(n-1) \\ \gamma(n-2) \\ \vdots \\ \gamma(1) \end{bmatrix}$$

Simplify to γ_n

$E(x_{n+1}x_2)$

$E(x_{n+1}x_n)$

$$\phi_n = \Gamma_n^{-1} \gamma_n$$

Cross covariance between x_n and x_k

Inverse of the covariance of the known function = Precision

Forecasting

Summary: Predictor

One-step-ahead predictor: $x_{n+1}^n = \sum_{k=1}^n \phi_k x_k = \boldsymbol{\phi}_n' \mathbf{x}$

$$\boldsymbol{\phi}_n = \Gamma_n^{-1} \boldsymbol{\gamma}_n, \quad P_{n+1}^n = \gamma(0) - \boldsymbol{\gamma}_n' \Gamma_n^{-1} \boldsymbol{\gamma}_n$$

m -step-ahead predictor: $x_{n+m}^n = \sum_{k=1}^n \phi_k x_k = \boldsymbol{\phi}_n' \mathbf{x}$

$$\boldsymbol{\phi}_n = \Gamma_n^{-1} \boldsymbol{\gamma}_n^{(m)}, \quad P_{n+m}^n = \gamma(0) - \boldsymbol{\gamma}_n^{(m)'} \Gamma_n^{-1} \boldsymbol{\gamma}_n^{(m)}$$

$$\phi_n = \Gamma_n^{-1} \gamma_n$$

Ex. 1 Prediction for an AR(2) process

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + w_t$$

Solutions

$t = 1$: No available observation.

Cross covariance between observation and prediction

$t = 2$: Available observation = x_1

Covariance of the observation

$$x_2^1 = \phi_{11} x_1 = \Gamma_1^{-1} \gamma_1 x_1 = \gamma(0)^{-1} \gamma(1) x_1 = \rho(1) x_1$$

$t = 3$: Available observation = x_1, x_2

$$\begin{aligned} x_3^2 &= \phi_{21} x_2 + \phi_{22} x_1 = (\Gamma_2^{-1} \gamma_2)' \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \gamma_2' \Gamma_2'^{-1} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= \begin{bmatrix} \gamma(3-1) \\ \gamma(3-2) \end{bmatrix}' \begin{bmatrix} \gamma(1-1) & \gamma(1-2) \\ \gamma(2-1) & \gamma(2-2) \end{bmatrix}^{-1} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

Ex. 1 Prediction for an AR(2) process (2)

$$\begin{aligned}
 x_3^2 &= \begin{bmatrix} \gamma(2) \\ \gamma(1) \end{bmatrix}' \begin{bmatrix} \gamma(0) & \gamma(1) \\ \gamma(1) & \gamma(0) \end{bmatrix}^{-1} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{\begin{bmatrix} \gamma(2) & \gamma(1) \end{bmatrix}}{\gamma(0)^2 - \gamma(1)^2} \begin{bmatrix} \gamma(0) & -\gamma(1) \\ -\gamma(1) & \gamma(0) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
 &= \frac{\begin{bmatrix} \rho(2) & \rho(1) \end{bmatrix}}{1 - \rho(1)^2} \begin{bmatrix} 1 & -\rho(1) \\ -\rho(1) & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}
 \end{aligned}$$

$\begin{bmatrix} \alpha c_{11} & \alpha c_{12} \\ \alpha c_{21} & \alpha c_{22} \end{bmatrix} = \alpha \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$

$\vdots$

Let's find $\rho(1)$ and $\rho(2)$.

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + w_t$$

$$\gamma(0) = E(x_t^2)$$

$$= E(\phi_1^2 x_{t-1}^2 + \phi_2^2 x_{t-2}^2 + w_t^2 + 2\phi_1\phi_2 x_{t-1}x_{t-2} + 2\phi_1 x_{t-1}w_t + 2\phi_2 x_{t-2}w_t)$$

$$= \phi_1^2 \gamma(0) + \phi_2^2 \gamma(0) + \sigma_w^2 + 2\phi_1\phi_2 \gamma(1) \tag{i}$$

Ex. 1 Prediction for an AR(2) process (3)

$$\begin{aligned}\gamma(1) &= E(x_t x_{t+1}) = E(x_t (\phi_1 x_t + \phi_2 x_{t-1} + w_{t+1})) \\ &= E(\phi_1 x_t^2 + \phi_2 x_t x_{t-1} + x_t w_{t+1})\end{aligned}$$

$$\gamma(1) = \phi_1 \gamma(0) + \phi_2 \gamma(1) \quad (\text{ii})$$

$$\phi_1 \gamma(0) - (1 - \phi_2) \gamma(1) = 0$$

From (i) and (ii), we get

$$(1 - \phi_1^2 - \phi_2^2) \gamma(0) - 2\phi_1 \phi_2 \gamma(1) = \sigma_w^2$$

$$\gamma(0) = \frac{(1 - \phi_2) \sigma_w^2}{(1 + \phi_2) \left((1 - \phi_2)^2 - \phi_1^2 \right)},$$

$$\begin{aligned}\gamma(1) &= \frac{\phi_1}{(1 - \phi_2)} \gamma(0) \\ \rho(1) &= \frac{\phi_1}{(1 - \phi_2)}\end{aligned}$$

Ex. 1 Prediction for an AR(2) process (4)

Case $h \geq 2$:

$$\gamma(h) = E(x_t x_{t+h}) = E(x_t (\phi_1 x_{t+h-1} + \phi_2 x_{t+h-2} + w_{t+h}))$$

$$= E(\phi_1 x_t x_{t+h-1} + \phi_2 x_t x_{t+h-2} + x_t w_{t+h})$$

$$= \phi_1 \gamma(h-1) + \phi_2 \gamma(h-2)$$

$$\gamma(2) = \phi_1 \gamma(1) + \phi_2 \gamma(0) = \phi_1 \frac{\phi_1}{(1 - \phi_2)} \gamma(0) + \phi_2 \gamma(0)$$

$$= \frac{\phi_1^2 - \phi_2^2 + \phi_2}{1 - \phi_2} \gamma(0)$$

$$\rho(2) = \frac{\phi_1^2 - \phi_2^2 + \phi_2}{1 - \phi_2}$$

$$\rho(1) = \frac{\phi_1}{1 - \phi_2}, \quad \rho(2) = \frac{(\phi_1^2 - \phi_2^2 + \phi_2)}{1 - \phi_2}$$

Ex. 1 Prediction for an AR(2) process (5)

$$x_2^1 = \phi_{11}x_1 = \rho(1)x_1 = \frac{\phi_1}{1 - \phi_2}x_1$$

$$x_3^2 = \phi_{21}x_2 + \phi_{22}x_1 = \frac{[\rho(2) \quad \rho(1)]}{1 - \rho(1)^2} \begin{bmatrix} 1 & -\rho(1) \\ -\rho(1) & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \frac{1}{1 - \rho(1)^2} [(\rho(2) - \rho(1)^2)x_1 + \rho(1)(1 - \rho(2))x_2]$$

$$= \frac{(1 - \phi_2)^2}{(1 - \phi_2)^2 - \phi_1^2} \left[\frac{\phi_2 \left((1 - \phi_2)^2 - \phi_1^2 \right)}{(1 - \phi_2)^2} x_1 + \frac{\phi_1 \left((1 - 2\phi_2 + \phi_2^2) - \phi_1^2 \right)}{(1 - \phi_2)^2} x_2 \right]$$

$$= \phi_2 x_1 + \phi_1 x_2$$

Predictor of AR(p) model

$$x_t = \boxed{\phi_1}x_{t-1} + \boxed{\phi_2}x_{t-2} + \cdots + \boxed{\phi_p}x_{t-p} + \boxed{w_t}$$

Independent to the past x

$$x_{n+1}^n = \sum_{k=1}^n \phi_{n,(n-k)} x_k$$

$$= \phi_{n,(n-1)}x_1 + \phi_{n,(n-2)}x_2 + \cdots + \phi_{n,2}x_{n-2} + \phi_{n,1}x_{n-1}$$

$$= \boxed{\phi_{n1}}x_{n-1} + \boxed{\phi_{n2}}x_{n-2} + \cdots + \boxed{\phi_{n,p}}x_{n+1-p} + \cdots + \phi_{n,n-2}x_2 + \phi_{n,n-1}x_1$$

Unnecessary information

$$= \phi_1x_{n-1} + \phi_2x_{n-2} + \cdots + \phi_px_{n-p+1}$$

Durbin-Levinson Algorithm

$$x_{n+1}^n = \phi_{n1}x_n + \phi_{n2}x_{n-1} + \cdots + \phi_{nn}x_1$$

1. Initial $\phi_{00} = 0, \quad P_1^0 = \gamma(0)$

2. Iteration

2.1 For $n \geq 1$,

$$\phi_{nn} = \frac{\rho(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(n-k)}{1 - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(k)}, \quad P_{n+1}^n = P_n^{n-1} (1 - \phi_{nn}^2)$$

2.2 For $n \geq 2$,

$$\phi_{nk} = \phi_{n-1,k} - \phi_{nn} \phi_{n-1,n-k}, \quad k = 1, 2, \dots, n-1$$

Ex. 2 Durbin-Levinson Algorithm

$$\begin{aligned}\phi_{nn} &= \frac{\rho(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(n-k)}{1 - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(k)} \\ \phi_{nk} &= \phi_{n-1,k} - \phi_{nn} \phi_{n-1,n-k}, n \geq 2 \\ P_{n+1}^n &= P_n^{n-1} (1 - \phi_{nn}^2)\end{aligned}$$

1. Initial $\phi_{00} = 0, \quad P_1^0 = \gamma(0)$

2. Iteration

$n = 1:$
$$\phi_{11} = \frac{\rho(1) - \sum_{k=1}^0 \phi_{0,k} \rho(1-k)}{1 - \sum_{k=1}^0 \phi_{0,k} \rho(k)} = \boxed{\rho(1)}$$

$$P_2^1 = P_1^0 (1 - \phi_{11}^2) = \gamma(0) (1 - \rho(1)^2)$$

$n = 2:$
$$\begin{aligned}\phi_{22} &= \frac{\rho(2) - \sum_{k=1}^1 \phi_{1,k} \rho(2-k)}{1 - \sum_{k=1}^1 \phi_{1,k} \rho(k)} = \frac{\rho(2) - \boxed{\phi_{11}} \rho(1)}{1 - \phi_{11} \rho(1)} \\ &= \frac{\rho(2) - \rho(1)^2}{1 - \rho(1)^2}\end{aligned}$$

Ex. 2 Durbin-Levinson Algorithm (2)

$$\phi_{11} = \rho(1), \quad P_2^1 = \gamma(0)(1 - \rho(1)^2), \quad \phi_{22} = \frac{\rho(2) - \rho(1)^2}{1 - \rho(1)^2}$$

$$\begin{aligned} k = 1: \quad \phi_{21} &= \phi_{11} - \phi_{22}\phi_{11} = \rho(1) - \frac{\rho(2) - \rho(1)^2}{1 - \rho(1)^2} \rho(1) \\ &= \frac{\rho(1)}{1 - \rho(1)^2} (1 - \rho(2)) \end{aligned}$$

$$P_3^2 = P_2^1(1 - \phi_{22}^2) = \gamma(0)(1 - \rho(1)^2)(1 - \phi_{22}^2)$$

Plug in the value from AR(2) process (Ex. 1), we have

$$\phi_{22} = \phi_2, \phi_{21} = \phi_1, P_3^2 = \sigma_w^2.$$

$$\begin{aligned} \phi_{nn} &= \frac{\rho(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(n-k)}{1 - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(k)} \\ \phi_{nk} &= \phi_{n-1,k} - \phi_{nn} \phi_{n-1,n-k}, n \geq 2 \\ P_{n+1}^n &= P_n^{n-1} (1 - \phi_{nn}^2) \end{aligned}$$

Ex. 2 Durbin-Levinson Algorithm (3)

From $n = 2$, we have $\phi_{22}, \phi_{21}, P_3^2$

$$n = 3: \quad \phi_{33} = \frac{\rho(3) - \sum_{k=1}^2 \phi_{2,k} \rho(3-k)}{1 - \sum_{k=1}^2 \phi_{2,k} \rho(k)} = \frac{\rho(3) - \phi_{21} \rho(2) - \phi_{22} \rho(1)}{1 - \phi_{21} \rho(1) - \phi_{22} \rho(2)}$$

$$k = 2: \quad \phi_{32} = \phi_{22} - \phi_{33} \phi_{21}$$

$$k = 1: \quad \phi_{31} = \phi_{21} - \phi_{33} \phi_{22}$$

$$P_4^3 = P_3^2 (1 - \phi_{33}^2)$$

$$\begin{aligned} \phi_{nn} &= \frac{\rho(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(n-k)}{1 - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(k)} \\ \phi_{nk} &= \phi_{n-1,k} - \phi_{nn} \phi_{n-1,n-k}, n \geq 2 \\ P_{n+1}^n &= P_n^{n-1} (1 - \phi_{nn}^2) \end{aligned}$$

Plug in the value from AR(2) process, we have
 $\phi_{33} = 0, \phi_{32} = \phi_{22} = \phi_2, \phi_{31} = \phi_{21} = \phi_1, P_4^3 = P_3^2.$

Ex.2 Durbin-Levinson Algorithm (4)

$$\text{For } n \geq 2, \quad \phi_{nk} = \begin{cases} \phi_1, & k = 1, \\ \phi_2, & k = 2, \\ 0, & k > 2 \end{cases} \quad P_{n+1}^n = \sigma_w^2$$

$$\begin{aligned} x_{n+1}^n &= \phi_{n1}x_n + \phi_{n2}x_{n-1} + \cdots + \phi_{nn}x_1 \\ &= \phi_1x_n + \phi_2x_{n-1} \end{aligned}$$

$$\text{AR(2) model: } x_{n+1} = \phi_1x_n + \phi_2x_{n-2} + w_t$$

$$\text{Error} = x_{n+1} - x_{n+1}^n = \text{noise} \text{ so the MSE is } E(w_t^2) = \sigma_w^2$$

PACF of stationary process

The PACF of a stationary process x_t can be obtained via Durbin-Levinson algorithm as ϕ_{nn} , for $n = 1, 2, \dots$

Innovation Algorithm

$$x_{n+1}^n = \theta_{n1}(x_n - x_n^{n-1}) + \theta_{n2}(\boxed{x_{n-1} - x_{n-1}^{n-2}}) + \cdots + \theta_{nn}(x_1 - x_1^0)$$

innovation

1. Initial $x_1^0 = 0, \quad P_1^0 = \gamma(0)$

2. Iteration: $n = 1, 2, \dots$

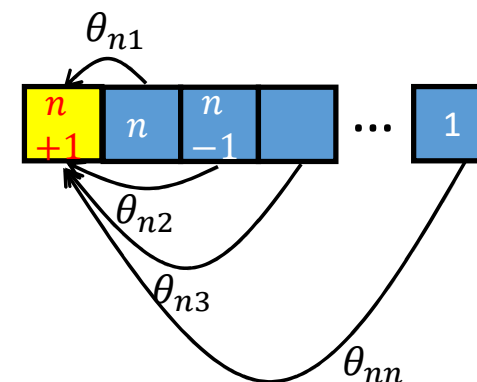
$$\theta_{n,n-j} = \frac{(\gamma(n-j) - \sum_{k=0}^{j-1} \theta_{j,j-k} \theta_{n,n-k} P_{k+1}^k)}{P_{j+1}^j}, j = 0, 1, \dots, n-1$$

Reduction of error from known innovation

$$P_{n+1}^n = \gamma(0) - \sum_{j=0}^{n-1} \boxed{\theta_{n,n-j}^2 P_{j+1}^j}$$

Idea of innovation algorithm

$$\begin{aligned}
 x_{n+1}^n &= \theta_{n1}(x_n - x_n^{n-1}) + \theta_{n2}(x_{n-1} - x_{n-1}^{n-2}) + \cdots + \theta_{nn}(x_1 - x_1^0) \\
 &= \theta_{n1}v_n + \theta_{n2}v_{n-1} + \cdots + \theta_{nn}v_1
 \end{aligned}$$



$$P_{n+1}^n = \gamma(0) - \sum_{j=0}^{n-1} \theta_{n,n-j}^2 P_{j+1}^j$$

$$\begin{aligned}
 &= \gamma(0) - \underbrace{\theta_{n1}^2 P_n^{n-1}}_{\text{Error reduction from } v_n} - \underbrace{\theta_{n2}^2 P_{n-1}^{n-2}}_{\text{Error reduction from } v_{n-1}} - \cdots - \underbrace{\theta_{nn}^2 P_1^0}_{\text{Error reduction from } v_1}
 \end{aligned}$$

M-step-ahead predictor



$$x_{n+m}^{n+m-1} = \sum_{j=1}^{n+m-1} \theta_{n+m-1,j} v_{n+m-j}$$

$$= \sum_{j=1}^{m-1} \theta_{n+m-1,j} v_{n+m-j} + \sum_{j=m}^{n+m-1} \theta_{n+m-1,j} v_{n+m-j}$$

$$P_{n+m}^{n+m-1} = \gamma(0) - \sum_{j=0}^{n+m-2} \theta_{n+m-1,n+m-1-j}^2 P_{j+1}^j$$

$$= \gamma(0) - \sum_{j=0}^{n-1} \theta_{n+m-1,n+m-1-j}^2 P_{j+1}^j - \sum_{j=n}^{n+m-2} \theta_{n+m-1,n+m-1-j}^2 P_{j+1}^j$$

Forecasting

M-step-ahead predictor (2)

$$\begin{aligned}
 P_{n+m-1}^n &= \gamma(0) - \sum_{j=0}^{n-1} \theta_{n+m-1, n+m-1-j}^2 P_{j+1}^j \\
 &= \gamma(0) - \theta_{n+m-1, n+m-1}^2 P_1^0 - \theta_{n+m-1, n+m-2}^2 P_2^1 - \dots - \theta_{n+m-1, m}^2 P_n^{n-1} \\
 &= \gamma(0) - \theta_{n+m-1, m}^2 P_n^{n-1} - \theta_{n+m-1, m+1}^2 P_{n-1}^{n-2} - \dots - \theta_{n+m-1, m+n-1}^2 P_1^0 \\
 P_{n+m-1}^n &= \gamma(0) - \sum_{j=m}^{n+m-1} \theta_{n+m-1, j}^2 P_{n+m-j}^{n+m-j-1}
 \end{aligned}$$

M-step-ahead predictor: Innovation algorithm

$$x_{n+m}^n = \sum_{j=m}^{n+m-1} \theta_{n+m-1,j} \left(x_{n+m-j} - x_{n+m-j}^{n+m-j-1} \right)$$

$$\theta_{n,n-j} = \frac{\left(\gamma(n-j) - \sum_{k=0}^{j-1} \theta_{j,j-k} \theta_{n,n-k} P_{k+1}^k \right)}{P_{j+1}^j}, j = 0, 1, \dots, n-1$$

$$P_{n+m-1}^n = \gamma(0) - \sum_{j=m}^{n+m-1} \theta_{n+m-1,j}^2 P_{n+m-j}^{n+m-j-1}$$

Ex. 3: Prediction for an MA(1)

$$x_t = w_t + \theta_1 w_{t-1} \quad \left\{ \gamma(h) = \begin{cases} (1 + \theta_1^2) \sigma_w^2, & h = 0, \\ \theta_1 \sigma_w^2, & h = 1, \\ 0, & h \geq 2 \end{cases} \right.$$

1. Initial $x_1^0 = 0, \quad P_1^0 = \gamma(0)$

2. Iteration:

$$n = 1, j = 0: \quad \theta_{11} = \frac{\gamma(1) - \sum_{k=0}^{-1} \theta_{0,-k} \theta_{1,1-k} P_{k+1}^k}{P_1^0} = \frac{\gamma(1)}{P_1^0}$$

$$P_2^1 = \gamma(0) - \sum_{j=0}^0 \theta_{1,1-j}^2 P_{j+1}^j = \gamma(0) - \theta_{11}^2 P_1^0 = \gamma(0) - \frac{\gamma(1)^2}{P_1^0}$$

$$\theta_{n,n-j} = \frac{(\gamma(n-j) - \sum_{k=0}^{j-1} \theta_{j,j-k} \theta_{n,n-k} P_{k+1}^k)}{P_{j+1}^j}$$

Forecasting

$$P_{n+1}^n = \gamma(0) - \sum_{j=0}^{n-1} \theta_{n,n-j}^2 P_{j+1}^j$$

Ex. 3: Prediction for an MA(1) (2)

$$\gamma(h) = \begin{cases} (1 + \theta_1^2)\sigma_w^2, & h = 0, \\ \theta_1\sigma_w^2, & h = 1, \\ 0, & h \geq 2 \end{cases}$$

$$n = 2, j = 0: \quad \theta_{22} = \frac{\gamma(2) - \sum_{k=0}^{-1} \theta_{0,-k} \theta_{2,2-k} P_{k+1}^k}{P_1^0} = \frac{\gamma(2)}{\gamma(0)} = 0$$

$$j = 1: \quad \theta_{21} = \frac{\gamma(1) - \sum_{k=0}^0 \theta_{1,1-k} \theta_{2,2-k} P_{k+1}^k}{P_2^1} = \frac{\gamma(1) - \theta_{11} \theta_{22} P_1^0}{P_2^1}$$

$$= \frac{\gamma(1)}{P_2^1}$$

$$P_3^2 = \gamma(0) - \sum_{j=0}^1 \theta_{2,2-j}^2 P_{j+1}^j = \gamma(0) - \theta_{22}^2 P_1^0 - \theta_{21}^2 P_2^1 = \gamma(0) - \frac{\gamma(1)^2}{P_2^1}$$

$$\theta_{n,n-j} = \frac{(\gamma(n-j) - \sum_{k=0}^{j-1} \theta_{j,j-k} \theta_{n,n-k} P_{k+1}^k)}{P_{j+1}^j}$$

Forecasting

$$P_{n+1}^n = \gamma(0) - \sum_{j=0}^{n-1} \theta_{n,n-j}^2 P_{j+1}^j$$

Ex. 3: Prediction for an MA(1) (3)

$$\gamma(h) = \begin{cases} (1 + \theta_1^2)\sigma_w^2, & h = 0, \\ \theta_1\sigma_w^2, & h = 1, \\ 0, & h \geq 2 \end{cases}$$

$$n = 3, j = 0: \quad \theta_{33} = \frac{\gamma(3) - \sum_{k=0}^{-1} \theta_{0,-k} \theta_{3,3-k} P_{k+1}^k}{P_1^0} = \frac{\gamma(3)}{\gamma(0)} = 0$$

$$j = 1: \quad \theta_{32} = \frac{\gamma(2) - \sum_{k=0}^0 \theta_{1,1-k} \theta_{3,3-k} P_{k+1}^k}{P_2^1} = \frac{\gamma(2) - \theta_{11} \theta_{33} P_1^0}{P_2^1} = 0$$

$$j = 2: \quad \theta_{31} = \frac{\gamma(1) - \sum_{k=0}^1 \theta_{2,2-k} \theta_{3,3-k} P_{k+1}^k}{P_3^2} \\ = \frac{\gamma(1) - \theta_{22} \theta_{33} P_1^0 - \theta_{21} \theta_{32} P_2^1}{P_3^2} = \frac{\gamma(1)}{P_3^2}$$

$$\theta_{n,n-j} = \frac{(\gamma(n-j) - \sum_{k=0}^{j-1} \theta_{j,j-k} \theta_{n,n-k} P_{k+1}^k)}{P_{j+1}^j}$$

Forecasting

$$P_{n+1}^n = \gamma(0) - \sum_{j=0}^{n-1} \theta_{n,n-j}^2 P_{j+1}^j$$

$$P_{n+1}^n = \gamma(0) - \sum_{j=0}^{n-1} \theta_{n,n-j}^2 P_{j+1}^j$$

Ex. 3: Prediction for an MA(1) (4)

$$\begin{aligned} n = 3, \quad P_4^3 &= \gamma(0) - \sum_{j=0}^2 \theta_{3,3-j}^2 P_{j+1}^j \\ &= \gamma(0) - \theta_{33}^2 P_1^0 - \theta_{32}^2 P_2^1 - \theta_{31}^2 P_1^0 = \gamma(0) - \frac{\gamma(1)^2}{P_3^2} \end{aligned}$$

$$\theta_{11} = \frac{\gamma(1)}{P_1^0}$$

$$\theta_{22} = 0$$

$$\theta_{21} = \frac{\gamma(1)}{P_2^1}$$

$$\theta_{33} = \theta_{32} = 0$$

$$\theta_{31} = \frac{\gamma(1)}{P_3^2}$$

$$P_2^1 = \gamma(0) - \frac{\gamma(1)^2}{P_1^0}$$

$$P_3^2 = \gamma(0) - \frac{\gamma(1)^2}{P_2^1}$$

$$P_4^3 = \gamma(0) - \frac{\gamma(1)^2}{P_3^2}$$

$$\theta_{nk} = \begin{cases} 0, & 1 < k < n \\ \frac{\gamma(1)}{P_n^{n-1}}, & k = 1 \end{cases}$$

$$\begin{aligned} P_{n+1}^n &= \gamma(0) - \gamma(1) \frac{\gamma(1)}{P_n^{n-1}} \\ &= \gamma(0) - \gamma(1) \theta_{n1} \end{aligned}$$

Ex. 3: Prediction for an MA(1) (5)

$$\theta_{nk} = \begin{cases} 0, & 1 < k < n \\ \frac{\gamma(1)}{p_n^{n-1}}, & k = 1 \end{cases}$$

$$\begin{aligned} x_{n+1}^n &= \sum_{k=1}^n \theta_{nk} (x_{n-k+1} - x_{n-k+1}^{n-k}) = \theta_{n1} (x_n - x_n^{n-1}) \\ &= \frac{\theta_1 \sigma_w^2}{p_n^{n-1}} (x_n - x_n^{n-1}) \end{aligned}$$

MA(1) model: $x_{n+1} = w_{n+1} + \theta_1 w_n$

Durbin-Levinson

$$x_{n+1}^n = \sum_{k=1}^n \phi_{nk} x_{n-k+1}$$

$$\phi_{00} = 0, \quad P_1^0 = \gamma(0)$$

$$\phi_{nn} = \frac{\rho(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(n-k)}{1 - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(k)}$$

$$\phi_{nk} = \phi_{n-1,k} - \phi_{nn} \phi_{n-1,n-k}$$

$$P_{n+1}^n = P_n^{n-1} (1 - \phi_{nn}^2)$$

VS

Iterative

$$x_{n+1}^n = \sum_{k=1}^n \theta_{nk} \left(x_{n-k+1} - \overset{\text{Innovation} \approx \text{noise}}{x_{n-k+1}^{n-k}} \right)$$

$$x_1^0 = 0, \quad P_1^0 = \gamma(0)$$

$$\theta_{n,n-j} = \frac{(\gamma(n-j) - \sum_{k=0}^{j-1} \theta_{j,j-k} \theta_{n,n-k} P_{k+1}^k)}{P_{j+1}^j}$$

$$P_{n+1}^n = \gamma(0) - \sum_{j=0}^{n-1} \theta_{n,n-j}^2 P_{j+1}^j$$

Forecasting ARMA process

$$x_{n+m}^n = E(x_{n+m} | \overbrace{x_n, x_{n-1}, \dots, x_1}^{\text{observations}})$$

ARMA(p, q) process:

$$\begin{aligned} x_{n+m} &= \sum_{k=1}^p \phi_k x_{n+m-k} + w_{n+m} + \sum_{l=1}^q \theta_l w_{n+m-l} \\ &= \sum_{k=1}^{\infty} -\pi_k x_{n+m-k} + w_{n+m} = \underbrace{\sum_{k=1}^{m-1} (\cdot)}_{\text{future}} + \underbrace{\sum_{k=m}^{n+m-1} (\cdot)}_{\text{observations}} + \underbrace{\sum_{k=n+m}^{\infty} (\cdot)}_{\text{unobservable}} + w_{n+m} \\ &= w_{n+m} + \sum_{l=1}^{\infty} \psi_l w_{n+m-l} = w_{n+m} + \underbrace{\sum_{l=1}^{m-1} (\cdot)}_{\text{Forecasting}} + \sum_{l=m}^{n+m-1} (\cdot) + \sum_{l=n+m}^{\infty} (\cdot) \end{aligned}$$

Linear MMSE predictor (2)

$$\begin{aligned}\tilde{x}_{n+m} &= E(x_{n+m} | x_n, x_{n-1}, \dots, x_1, x_0, x_{-1}, \dots) \\ &= E\left(\sum_{l=0}^{\infty} \psi_l w_{n+m-l} \mid x_n, x_{n-1}, \dots, x_1, x_0, x_{-1}, \dots\right), \psi_0 = 1\end{aligned}$$

$$\text{Let } \tilde{w}_{n+m} = E(w_{n+m} | x_n, x_{n-1}, \dots, x_1, x_0, x_{-1}, \dots) = \begin{cases} 0, & m > 0 \\ w_{n+m}, & m \leq 0 \end{cases}$$

$$\tilde{x}_{n+m} = \sum_{l=0}^{\infty} \psi_l \tilde{w}_{n+m-l} = \underbrace{\sum_{l=0}^{m-1} \psi_l \tilde{w}_{n+m-l}}_{\text{Forecasting}} + \sum_{l=m}^{\infty} \psi_l w_{n+m-l}$$

Linear MMSE predictor: MSE

Missing (future) information

$$x_{n+m} = \sum_{l=0}^{m-1} \psi_l w_{n+m-l} + \sum_{l=m}^{n+m-1} \psi_l w_{n+m-l} + \sum_{l=n+m}^{\infty} \psi_l w_{n+m-l}$$

$$\tilde{x}_{n+m} = \sum_{l=m}^{\infty} \psi_l w_{n+m-l}$$

$$\text{error} = x_{n+m} - \tilde{x}_{n+m} = \sum_{l=0}^{m-1} \psi_l w_{n+m-l}$$

$$\text{MSE} = E(x_{n+m} - \tilde{x}_{n+m})^2 = \sigma_w^2 \sum_{l=0}^{m-1} \psi_l^2$$

ARMA forecast: large m

$$\tilde{x}_{n+m} = \sum_{l=m}^{\infty} \psi_l w_{n+m-l}$$

$$\text{If } E(x) = \mu_x, \quad (\tilde{x}_{n+m} - \mu_x) = \sum_{l=m}^{\infty} \psi_l w_{n+m-l}$$

$$\tilde{x}_{n+m} = \mu_x + \sum_{l=m}^{\infty} \psi_l w_{n+m-l}$$

If $m \rightarrow \infty$, $\psi_m \rightarrow 0$ and $\tilde{x}_{n+m} \rightarrow \mu_x$

$$\text{MSE} = E(x_{n+m} - \tilde{x}_{n+m})^2 = E(x_{n+m} - \mu_x)^2 = \gamma(0) = \sigma_x^2$$

ARMA forecast: truncated predictor

$$\begin{aligned}
 x_{n+m} &= w_{n+m} - \sum_{k=1}^{m-1} \pi_k x_{n+m-k} - \sum_{l=m}^{n+m-1} \pi_l x_{n+m-l} - \sum_{l=n+m}^{\infty} \pi_l x_{n+m-l} \\
 \tilde{x}_{n+m} &= \tilde{w}_{n+m} - \sum_{k=1}^{m-1} \pi_k \boxed{\tilde{x}_{n+m-k}}^{\text{predictable}} - \sum_{l=m}^{n+m-1} \pi_l \boxed{\tilde{x}_{n+m-l}}^{\text{observable}} - \sum_{l=n+m}^{\infty} \pi_l \boxed{\tilde{x}_{n+m-l}}^{\text{unobservable}}
 \end{aligned}$$

$$\tilde{x}_{n+m}^n = - \sum_{k=1}^{m-1} \pi_k \tilde{x}_{n+m-k}^n - \sum_{l=m}^{n+m-1} \pi_l x_{n+m-l}, \text{MSE} \approx \sigma_w^2 \sum_{l=0}^{m-1} \psi_l^2$$

Truncated prediction for ARMA

$$x_{n+m} = \phi_1 x_{n+m-1} + \cdots + \phi_p x_{n+m-p} + w_{n+m} + \theta_1 w_{n+m-1} + \cdots + \theta_q w_{n+m-q}$$

Find conditional expectation.

$$\tilde{x}_{n+m}^n = \phi_1 \tilde{x}_{n+m-1}^n + \cdots + \phi_p \tilde{x}_{n+m-p}^n + \theta_1 \tilde{w}_{n+m-1}^n + \cdots + \theta_q \tilde{w}_{n+m-q}^n$$

$$\tilde{x}_t^n = E(x_t | x_1, x_2, \dots, x_n) = \begin{cases} 0, & t \leq 0 \\ x_t & 1 \leq t \leq n \\ \sum_{k=1}^n \alpha_k x_k, & n < t \end{cases}$$

$$\tilde{w}_t^n = \begin{cases} 0, & t \leq 0, t > n \\ \phi(B)x_t - \theta_1 \tilde{w}_{t-1}^n - \cdots - \theta_q \tilde{w}_{t-q}^n, & 1 \leq t \leq n \end{cases}$$

Truncated prediction error

Forecasting

Ex. 4: Forecasting ARMA(1,1) Series

$$x_{n+1} = \phi_1 x_n + w_{n+1} + \theta_1 w_n$$

One-step-ahead predictor: $\tilde{x}_{n+1}^n = \phi_1 x_n + 0 + \theta_1 \tilde{w}_n^n$

$$\tilde{w}_n^n = x_n - \phi_1 x_{n-1} + \theta_1 \tilde{w}_{n-1}^n$$

m -step-ahead predictor ($m > 1$): $\tilde{x}_{n+m}^n = \phi_1 \tilde{x}_{n+m-1}^n + 0 + 0$

Prediction error: $P_{n+m}^n = \sigma_w^2 \sum_{l=0}^{m-1} \psi_l^2$

Backcasting

Find the value in the past such as x_{1-m} , $m = 1, 2, \dots$

$$x_{1-m}^n = \phi_{n1}^{(m)} x_1 + \phi_{n2}^{(m)} x_2 + \dots + \phi_{nn}^{(m)} x_n$$

Use forecasting method but reverse the time direction.

$$\boldsymbol{\phi}_n^{(m)} = \begin{bmatrix} \phi_{nn}^{(m)} \\ \phi_{n,n-1}^{(m)} \\ \vdots \\ \phi_{n1}^{(m)} \end{bmatrix} = \Gamma_n^{-1} \gamma_n^m$$